

Department of Computer Science and Engineering
Jashore University of Science & Technology
3rd year, 1st Semester, Final Exam-2025 (Session: 2021-22)

Course Code: CSE 3101

Course Title: Operating System and System Programming

Total Time: 3.00 hours

Total Marks: 72.0

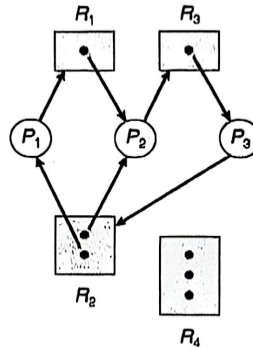
(Note: Answer any six set of questions from eight set of questions)

- ✓ 1. a) Define Operating System. Explain about computer system structure. 1+3
b) Why do some systems store the operating system in firmware, while others store it on disk? 2
c) What is interrupt? How can we handle interrupt? Describe. 4
d) Differentiate between client server computing and peer to peer computing. 2
2. a) Explain the services provided by operating system. 2
b) What are the two essential parts of a process? How is a process different from a program? 3
c) Describe process control block with proper figure. 4
d) Direct memory access is used for high-speed I/O devices in order to avoid increasing the CPU's execution load. 3
 ✓ i) How does the CPU interface with the device to coordinate the transfer?
 ii) The CPU is allowed to execute other programs while the DMA controller is transferring data. Does this process interfere with the execution of the user programs? If so, describe what forms of interference are caused.
- ✓ 3. a) Draw storage device hierarchy. 2
b) What do you know about critical section problem? How we can solve it? Describe. 2+2
c) Depict the control flow of system calls. Do you agree/disagree that the system calls are time consuming? Give your statement. 4
d) Mention some advantages of multithreaded programming. 2
- ✓ 4. a) Describe necessary and sufficient conditions to occur deadlock. 4
b) Consider the following set of process with CPU burst time. 5

Process	Burst Time
P1	10
P2	29
P3	3
P4	7
P5	12

- i) Consider the FCFS, SJF Non-Preemptive and Round Robin (Quantum=10) scheduling algorithms and calculate average waiting time to find which algorithm would give the minimum average waiting time.
- ✓ c) Shortly explain the scheduling criteria for CPU scheduling algorithms. 3
- ✓ 5. a) Consider a computer system that runs 5,000 jobs per month and has no deadlock-prevention or deadlock-avoidance scheme. Deadlocks occur about twice per month, and the operator must terminate and rerun about ten jobs per deadlock. Each job is worth about two dollars (in CPU time), and the jobs terminated tend to be about half done when they are aborted. 4
A systems programmer has estimated that a deadlock-avoidance algorithm (like the banker's algorithm) could be installed in the system with an increase of about 10 percent in the average execution time per job. Since the machine currently has 30 percent idle time, all 5,000 jobs per month could still be run, although turnaround time would increase by about 20 percent on average.
 ✓ i) What are the arguments for installing the deadlock-avoidance algorithm?
 ✓ ii) What are the arguments against installing the deadlock-avoidance algorithm?

- b) How we can recover from deadlock? Shortly describe. 2
- c) Consider $\alpha = 80\%$, $\epsilon = 20\text{ns}$ for TLB search, 100ns for memory access. Find out effective access time. 2
- d) What do you know about resource allocation graph? Consider the following graph and find out if there is any deadlock or not. Justify your answer. 1+3



6. a) Consider the following snapshot of a system: 4

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

- a) What is the content of the matrix Need? 4
- b) If a request from process P1 arrives for (1,4,6,0), can the request be granted immediately? 1+3
- b) Define paging. Shortly describe address translation scheme with suitable example. 4
- c) Design a multithreaded program to count and print the total number of words in two files. Give two use case scenarios for clarification. 4
7. a) Compare between physical address space and logical address space. 2
- b) What is fragmentation? Consider the following segment table: 5

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	58096
4	1952	

What are the physical addresses for the following logical addresses?

i) 0, 430	ii) 1, 10
iii) 2, 500	iv) 3, 400

- c) Define address binding. Explain multistep processing of a user program highlighting the different stages of address bindings with figure. 1+4
8. a) Consider the following reference string 1,2,3,2,5,6,3,4,6,3,4,6,3,7,3,1,5,3,6,3,4,2,4,3,4,5,1 and assume that the memory size is four frames. Calculate the number of page fault, page hit, and page miss ratio using optimal page replacement algorithm. 5
- b) What is Belady's Anomaly? Describe with proper example. 3
- c) Explain demand paging with proper figure. 4